

Arjan Faber

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Engineer specializing in real-time systems, robotics, and AI/ML for high-performance racing environments. Experienced in C++ and Python development, high-performance computing, and ROS-based simulation. Supported Formula 2 and Formula 3 racing operations, developing ML-driven tools and data systems to enhance performance, reliability, and decision-making.

EDUCATION

University of Edinburgh

Master of Science in Data Science

Edinburgh, United Kingdom

September 2023 - August 2025

- Dissertation at School of Informatics. Topic: "*Verifying the upper confidence bound (UCB) algorithm in Isabelle/HOL*".
- GPA (award): 68.5% (high merit).
- Courses: High Performance Computing, Threaded Programming, Data Science, Machine Learning, Numerical Algorithms.

Politecnico di Milano

Extracurricular graduate course in Mathematical Engineering

Milan, Italy

September 2024 - December 2024

- Course name: Computational Fluid Dynamics.
- Curriculum: Numerical methods for fluid dynamics, Navier-Stokes equations and discretization techniques.

University of Groningen

Bachelor of Science in Econometrics & Operations Research (EOR)

Groningen, Netherlands

January 2016 - July 2023

- Thesis topic : "*A comparison of machine learning methodologies for proxying CDS spreads*", awarded 8.0/10.
- Core statistics & econometrics courses GPA: 7.6/10 (70 ECTS). Overall GPA 6.7/10 (180 ECTS).
- Courses: Linear Algebra, Multivariate Calculus, Difference & Differential Equations, Programming in C++, Stochastic Models, Econometrics, Operations Research.

WORK EXPERIENCE

Van Amersfoort Racing B.V.

Artificial Intelligence (AI) & IT Support Engineer

Zeewolde, Netherlands

February 2026 – Present

- Developing a regulations LLM chat bot for the racing teams and a Meta Llama 3 Large Language Model (LLM) solution in Python to automate conversion of XLSM files to ICS format within Outlook, supporting the traveling coordinators scheduling efficiency for race weekends across Formula 2, Formula 3 and test teams.
- Sole in-house IT administrator responsible for teams' laptops and IT support, including installation, troubleshooting, SharePoint permissions, and server infrastructure visualization.
- Acting as the main internal point of contact and advocate for all AI and IT initiatives within the company.

TomTom

Statistical Data Engineering Intern

Amsterdam, Netherlands

June 2025 – January 2026

- Supporting global traffic teams with conducting thorough exploratory data analysis using Python and SQL.
- Developed and deployed an automated system that implemented Machine Learning (ML) techniques (**GMM, K-Means**) to evaluate, rank, and report the performance of third-party feed providers for high quality maps.
- Developing a large-scale language translator model implementing **LLMs** for drive test assistance.

Human Media Interaction (HMI) Group, University of Twente

Machine Learning Research Engineer

Enschede, Netherlands

January 2024 – July 2024

- Implemented **deep Q-learning** (DQN) and **inverse reinforcement learning** (MaxEnt IRL) to enable Kuka robots to acquire social behaviors in real-time, adapting to human feedback through multimodal cues.
- Led a data collection study using advanced sensors (LiDAR, RealSense cameras, Kinect) to gather > 25GB of interaction data. Implemented **OpenPose** and **CUDA** in **ROS Noetic** including **ROS2** bridge to process body and facial expressions for the robot's ability to interpret and respond to human behaviors.

Further details: <https://harmony-eu.org/>

ACADEMIC RESEARCH EXPERIENCE

School of Informatics, University of Edinburgh

Graduate researcher

Edinburgh, United Kingdom

February 2025 - August 2025

Dissertation topic: “*Verifying the upper confidence bound algorithm (UCB) in Isabelle/HOL*”. Officially published in the **Archive of Formal Proofs (AFP)** (public link: <https://www.isa-afp.org/entries/Discrete-UCB.html>). Key focus areas of my dissertation:

- Formalizes measure-theoretic probability in Isabelle/HOL to verify stochastic decision-making algorithms.
- Verifies concentration inequalities and tail bounds for UCB’s regret guarantees.
- Expands formal verification to reinforcement learning models.

Department of Economics, Econometrics and Finance, University of Groningen

Groningen, Netherlands

Undergraduate thesis student

September 2020 – January 2021

Thesis topic: “*A comparison of machine learning methodologies for proxying CDS spreads*”. Achievements:

- Transformed a single hidden layer Artificial Neural Network (ANN) into a **Bayesian Neural Network (BNN)** by considering the idea of weights and biases uncertainty introduced by Blundell et al. (2015).
- Introduced **local interpretable model-agnostic explanations (LIME)** to improve interpretability of the methodologies.

Keywords: Tensorflow, ProbFlow, Scikit-Learn, LIME.

EXTRACURRICULAR ACTIVITIES

Econometric Study Association VESTING

Groningen, Netherlands

Committee chairman & external affairs officer

September 2017 – May 2019

Co-responsible for the acquisition of foreign commercial partners for a study trip to Taipei City, Taiwan. Established partnership with Taiwan Stock Exchange, Deloitte, Willis Towers Watson and National Taiwan University.

Keywords: Team work, Communication skills, Cross-cultural.

SKILLS, ACTIVITIES & INTERESTS

Languages: Native in Dutch, fluent in English, basic German (reading).

IT Skills: Python, C, C++20 (memory management, Clang, Google Test), MATLAB, R, ReactJS, ROS1/ROS2, OpenMP, MPI, Linux, LaTeX, SLURM, ARCHER/ARCHER2 (HPC, job scheduling, resource management).

Interests: Jazz enthusiast (guitar), Formula 1, traveling (Asia, Scandinavia).